

Walk-unbiased doubly robust graph propagation: provably fair, transparent, accountable and private recommendation from exposure-biased logs

Abstract

Graph-based recommenders propagate preferences along multi-hop walks of a logged user–item graph. That graph has been filtered by the platform’s own exposure policy, so propagation amplifies exposure bias. Existing corrections re-weight edges by inverse propensities or impute missing edges with doubly robust (DR) estimates. We show that both remain biased beyond one hop: a walk that revisits an edge squares its estimate, which inflates exactly the rarely exposed edges that debiasing targets. The resulting bias grows with the inverse of the propensity clip. We propose DRUP (Doubly Robust, walk-Unbiased Propagation), which replaces every repeated edge by its idempotent unbiased counterpart. Through Möbius inversion over the coincidences of walk indices, DRUP is exactly unbiased and edge-wise doubly robust for propagation over the counterfactual full-exposure graph at any number of hops. It costs no more than the uncorrected operator and needs no training. The same structure yields provable properties beyond accuracy: variance and concentration bounds; exposure invariance as a causal item-fairness guarantee; exact Shapley attributions and optimal counterfactual explanations; closed-form certificates against injected profiles; joint differential privacy through a single public item operator; and exposure-capped

allocation with a causal utility guarantee. On unbiased test data from Coat, Yahoo! R3 and KuaiRec, DRUP significantly outperforms tuned MF, Light-GCN, NAVIP and DR baselines on Yahoo! R3 and KuaiRec, and it ties with the best trained model on Coat. Every guarantee is verified empirically. On very sparse logs, exposure invariance needs a small clip and costs accuracy.

Keywords: Causal recommendation, Graph neural networks, Doubly robust estimation, Exposure bias, Fairness, Explainability, Differential privacy

1. Introduction

Graph-based recommender systems (GBRs) represent users and items as nodes and logged interactions as edges. They score a user-item pair by aggregating signals along multi-hop walks (He et al., 2020; Shen et al., 2021). The logged graph, however, is not the preference graph. An edge exists only if the platform *exposed* the item and the user then responded. A recent survey of causal learning in GBRs (Yu et al., 2026) traces four open challenges to this gap: out-of-distribution generalisation, dynamic preferences, fairness and interpretability. All four arise because message passing propagates correlations of the observational graph rather than causal preference structure. The same survey lists the lack of theoretical guarantees, scalability and privacy as open directions.

Two families of fixes exist. *Loss-level* debiasing (IPS, DR) reweights the training objective (Schnabel et al., 2016; Wang et al., 2019; Saito et al., 2020) but leaves the propagation operator biased. *Propagation-level* debiasing reweights edges by inverse propensities during neighbour aggregation (Kim et al., 2022; Islam et al., 2026). We show that the second family

is itself biased for multi-hop propagation. The root cause is elementary. The edge estimate $W_e = O_e Y_e / p_e$ is unbiased for the potential outcome Y_e , but a walk that traverses e twice contributes W_e^2 , and $\mathbb{E}[W_e^2] = Y_e / p_e \neq Y_e$. Walks with repeated edges are not rare corner cases. They include every back-and-forth step $u \rightarrow j \rightarrow u$ that makes LightGCN self-reinforcing. Their bias is largest precisely where the propensity is smallest.

Contributions.

1. We identify the *repeated-walk bias* of inverse-propensity and DR graph propagation. We give its exact form and a lower bound that is unbounded as propensities vanish (Theorem 3).
2. We propose DRUP. Using $Y_e^2 = Y_e$, DRUP replaces each repeated edge by its idempotent unbiased counterpart. The resulting 3-hop operator is exactly unbiased and edge-wise doubly robust for propagation over the counterfactual full-exposure graph (Theorem 1). It has the cost of one item Gram matrix, needs no training, and factorises into a public item operator plus per-user local corrections. The correction extends exactly to any number of hops (Theorem 2).
3. We prove variance, clipping-bias and concentration bounds (Theorem 4).
4. We show that the same structure yields provable *FAT+P* properties: causal item fairness via exposure invariance (Theorem 5); exact attributions and optimal counterfactual explanations (Proposition 6, Theorem 7); certified robustness to injected profiles (Theorem 8); joint differential privacy (Theorem 9); and exposure-capped allocation with a causal utility guarantee (Theorem 10).

5. We evaluate DRUP on Coat (Schnabel et al., 2016) and Yahoo! R3 (Marlin and Zemel, 2009), whose test sets are random samples, and on KuaiRec (Gao et al., 2022), whose test matrix is fully observed. We compare it with trained and training-free baselines and verify each guarantee: by Monte Carlo, by a real-data intervention on exposure, by brute force, and against realised attacks.

2. Related work

Causal and debiased recommendation. Surveys organise the biases of recommendation data and their remedies (Chen et al., 2023) and the fairness notions of recommenders (Wang et al., 2023). Inverse propensity scoring (Schnabel et al., 2016) and doubly robust learning (Wang et al., 2019) are the standard tools for MNAR feedback, and a small unbiased sample can be used to learn debiasing parameters (Chen et al., 2021). Causal embeddings disentangle interest from conformity (Zheng et al., 2021). The survey of Yu et al. (2026) organises causal GBRs by challenge. None of these methods analyses the bias of the *propagation operator* beyond one hop.

Debiased message passing. NAVIP (Kim et al., 2022) applies inverse-propensity weights with Laplacian normalisation inside LightGCN aggregation. DPAA (Islam et al., 2026) learns popularity-aware aggregation weights. Both are unbiased for a single aggregation step at most. Theorem 3 shows that stacking such steps reintroduces a bias of order $1/p$.

Linear graph filters. LightGCN is a polynomial filter of the normalised adjacency, and closed-form filters are competitive with it (Shen et al., 2021).

DRUP is a closed-form filter on a debiased operator. Its linearity is what makes exact explanations, certificates and DP releases possible.

Explanations, robustness and privacy. Explainable recommendation is surveyed by Zhang and Chen (2020). GNNExplainer (Ying et al., 2019) and CF-GNNExplainer (Lucic et al., 2022) approximate explanations by optimising on a surrogate. PORE (Jia et al., 2023) certifies recommenders against poisoning by ensembling. Differentially private recommenders release noisy item covariances (McSherry and Mironov, 2009). DRUP gives exact explanations, closed-form certificates and a joint-DP guarantee (Hsu et al., 2014) for one and the same scoring rule.

3. Problem setting

Users $u \in [m]$, items $i \in [n]$ and pairs $e = (u, i)$ carry a binary potential outcome Y_e and an exposure $O_e \sim \text{Bernoulli}(p_e)$. The log reveals O and $O \odot Y$.

Assumption 1 (Unconfoundedness). *Given the propensities, the exposures O_e are mutually independent and independent of Y .*

Assumption 2 (Fixed nuisances). *The propensity estimate $\hat{p}$, the imputation $\hat{Y} \in [0, 1]^{m \times n}$ and the edge weights C do not depend on O . This holds with cross-fitting or with nuisances estimated on a vetted historical log. We use $C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$ with degrees taken from $\hat{Y}$.*

With $\tilde{Y} = C \odot Y$, the target is propagation over the full-exposure graph:

$$F^* = a \tilde{Y} + b T^*, \quad T^* = \tilde{Y} \tilde{Y}^\top \tilde{Y}. \quad (1)$$

This is the linear LightGCN / graph-filter score one would compute if every item had been shown to every user.

4. DRUP

Edge estimates. With clipped propensities $\bar{p} = \max(\hat{p}, \tau)$,

$$W_e^{\text{IPS}} = \frac{O_e Y_e}{\bar{p}_e}, \quad W_e^{\text{DR}} = \hat{Y}_e + \frac{O_e(Y_e - \hat{Y}_e)}{\bar{p}_e}. \quad (2)$$

Both have mean Y_e when $\bar{p} = p$.

Idempotent walk correction. Let $\tilde{W} = C \odot W$ and $\Delta = \tilde{W} \odot \tilde{W} - C \odot C \odot W$, with row sums r and column sums κ . The corrected 3-hop operator is

$$\hat{T} = \tilde{W} \tilde{W}^\top \tilde{W} - \tilde{W} \odot (r \mathbf{1}^\top - \Delta) - \tilde{W} \odot (\mathbf{1} \kappa^\top - \Delta) - (\tilde{W}^{\odot 3} - C^{\odot 3} \odot W). \quad (3)$$

The three corrections handle walks $u-j-v-i$ with $v = u$, with $j = i$, and with both. In each case every repeated factor $\tilde{W}_e^k$ is replaced by $C_e^k W_e$. The DRUP score is $s = a\tilde{W} + b\hat{T}$ with global constants a, b .

Public-operator form. Define the diagonal-corrected item Gram matrix $G = \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$. Then

$$\hat{T}_u = \tilde{w}_u G - \tilde{w}_u \odot (r_u \mathbf{1}^\top - 2\delta_u) - (\tilde{w}_u^{\odot 3} - c_u^{\odot 3} \odot w_u), \quad (4)$$

where $\delta_u = \tilde{w}_u^{\odot 2} - c_u^{\odot 2} \odot w_u$. Equation (4) matches (3) to machine precision. G is the only object shared across users (Algorithm 1).

Control-variate family. More generally, $W_e^\lambda = O_e Y_e / \bar{p}_e + \lambda(\hat{Y}_e - O_e \hat{Y}_e / \bar{p}_e)$ is unbiased for every $\lambda \in [0, 1]$ under correct propensities. It interpolates between IPS ($\lambda = 0$) and DR ($\lambda = 1$, the only member that is also robust to propensity errors). Its variance is $\frac{1-p}{p}(Y - \lambda \hat{Y})^2$. All results below hold for every member, and λ is selected on validation data.

Algorithm 1 DRUP (training-free)

Require: $\log(O, O \odot Y)$; hyper-parameters $\tau, \alpha, \lambda, b/a$

- 1: fit propensities $\hat{p}$ (logistic exposure model) and imputation $\hat{Y}$ (IPS-weighted additive ridge, optionally plus an IPS-weighted low-rank ALS residual)
 - 2: $\bar{p} \leftarrow \max(\hat{p}, \tau)$; $W \leftarrow \hat{Y} + O \odot (Y - \hat{Y})/\bar{p}$
 - 3: $C \leftarrow d_u^{-\alpha} d_i^{-(1-\alpha)}$ with d from $\hat{Y}$
 - 4: $G \leftarrow \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$ $\triangleright$ public operator
 - 5: **for** each user u (locally) **do**
 - 6: $\hat{T}_u \leftarrow \text{Eq. (4)}$; $s_u \leftarrow a \tilde{w}_u + b \hat{T}_u$; rank unexposed items by s_u
 - 7: **end for**
-

Imputation. Theorem 1 holds for any fixed $\hat{Y}$, so the imputation is a free design choice. Its accuracy enters only the variance (Theorem 4) and the clipping bias. We use either an additive model $\mu + b_u + b_i$ fitted by IPS-weighted ridge regression, or that model plus a rank-32 residual $UV^\top$ fitted by IPS-weighted alternating least squares on the logged pairs. The additive imputation gives every user the same item order on unexposed candidates, which makes the one-hop DR term non-personalised. The low-rank imputation removes this limitation.

Exposure-capped re-ranking. To control how concentrated the top- K lists are, the scores can be passed to an exposure-capped allocation (Theorem 10, solved exactly by min-cost flow).

Complexity. One $n \times n$ Gram product and $O(mn)$ reductions, i.e., the cost of the uncorrected operator. On KuaiRec (7,176 users, 10,728 items, 10.3M

exposures) one configuration takes under a minute on four CPU cores.

5. Theory

Full proofs are in Appendix Appendix A. Let $\varepsilon_e = |Y_e - \hat{Y}_e|$.

Theorem 1 (Edge-wise double robustness). *Under Assumptions 1–2, let $\delta_e = (p_e/\bar{p}_e - 1)(Y_e - \hat{Y}_e)$. Then $\mathbb{E}[\hat{T}_{ui}]$ equals T_{ui}^* with every Y_e replaced by $Y_e + \delta_e$. In particular, if every edge has a correct propensity or a correct imputation, then $\mathbb{E}[s] = F^*$.*

Theorem 2 (Any number of hops). *For odd $K = 2r + 1$, encode the coincidences of the index variables of a walk $U_0 - J_1 - \dots - U_r - J_{r+1}$ as a pair π of set partitions. π fixes the multiplicity k_e of every distinct edge. The estimator*

$$\hat{T}^{(K)} = \tilde{W}(\tilde{W}^\top \tilde{W})^r + \sum_{\pi: \exists k_e > 1} \sum_{\sigma \geq \pi} \mu(\pi, \sigma) (g_\pi^{\text{corr}}(\sigma) - g_\pi^{\text{plug}}(\sigma))$$

is exactly unbiased for $(\tilde{Y}\tilde{Y}^\top)^r \tilde{Y}$. Here $g_\pi(\sigma)$ is an unconstrained tensor contraction of $\prod_e C_e^{k_e} W_e$ (corrected) or $\prod_e \tilde{W}_e^{k_e}$ (plug-in) over the blocks of σ , and μ is the Möbius function of the partition lattice.

The number of contractions is 5, 99 and 2,790 for $K = 3, 5, 7$. For $K = 3$ the construction reduces to Eq. (3).

Theorem 3 (Repeated-walk bias of uncorrected propagation). *With correct propensities and a candidate (u, i) with $O_{ui} = 0$, the uncorrected DR propagation $\tilde{W}\tilde{W}^\top \tilde{W}$ has conditional bias*

$$C_{ui} \hat{Y}_{ui} (R_u + K_i) - C_{ui}^3 \hat{Y}_{ui} (1 - \hat{Y}_{ui}^2), \quad K_i = \sum_{v \neq u} C_{vi}^2 \frac{1 - p_{vi}}{p_{vi}} \varepsilon_{vi}^2,$$

and R_u is the analogous sum over the user's own edges. The within-user ranking bias $C_{ui}\hat{Y}_{ui}K_i$ satisfies $K_i \geq (1/\max_v p_{vi} - 1)\sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$. It therefore diverges as the item's propensities vanish, and it is $\Omega(1/\tau)$ under clipping. DRUP's bias is zero.

Theorem 4 (Variance, clipping and concentration). *Let $\Lambda_e(u, i)$ be the C -weighted mass of the walks from u to i through edge e , and let $B = \max(1, 1/\tau)$. Then (a) $\text{Var}(W_e) \leq \varepsilon_e^2/\tau$; (b) $\text{Var}(\hat{T}_{ui}) \leq \frac{(1+1/\tau)^2}{\tau} \sum_e \varepsilon_e^2 \Lambda_e(u, i)^2$; (c) $|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq 3\bar{\delta}(1 + \bar{\delta})^2 \sum_{j,v} C_{uj}C_{vj}C_{vi}$ with $\bar{\delta} = \max_e(1 - p_e/\tau)_+ \varepsilon_e$; (d) $\Pr(|\hat{T}_{ui} - \mathbb{E}\hat{T}_{ui}| \geq t) \leq 2\exp(-2t^2/\sum_e c_e^2)$ with $c_e = \varepsilon_e B^2 \Lambda_e(u, i)/\tau$.*

The variance scales with the imputation error ε^2 , not with Y^2 as for IPS, and the clip τ trades variance against a clipping bias that vanishes when $\hat{Y}$ is accurate.

Definition 1 (Exposure elasticity). For an intervention $do(p_i \leftarrow \pi p_i)$ on item i 's exposure mechanism, $\eta_i = \partial \log \mathbb{E}[s_{ui}]/\partial \log \pi$.

Theorem 5 (Exposure invariance: causal item fairness). *Under Assumptions 1–2 and $\pi p \geq \tau$, DRUP has $\eta_i = 0$, so items with equal potential outcomes receive equal expected scores whatever their exposure history. Logged-graph propagation has $\eta_i \geq 1$ (popularity amplification). Uncorrected DR propagation has $\eta_i < 0$ whenever $\hat{Y}_{ui}K_i > 0$ (over-correction).*

Proposition 6 (Affine in the user's own data). *Fix u and a candidate i with $O_{ui} = 0$, and let $G^{(-u)}$ be G without u 's own contribution. Then $s_{ui} = \text{const} + \sum_{j \neq i} (c_{uj}G_{ji}^{(-u)} + c_{ui}w_{ui}c_{uj}^2)w_{uj}$. Hence un-logging interaction j changes s_{ui} by exactly $\phi_j = a_j(w_{uj} - \hat{Y}_{uj})$. Removal effects add up over any subset, the ϕ_j are the exact Shapley values, and completeness holds exactly.*

Theorem 7 (Optimal minimal counterfactual explanation). *For candidates i, k with $s_{ui} > s_{uk}$, sorting $g_j = \phi_j^{(i)} - \phi_j^{(k)}$ in decreasing order and removing interactions until the cumulative sum exceeds the margin yields a minimum-cardinality set whose removal makes k outrank i . If the positive g_j do not sum above the margin, no such set exists. The cost is $O(n + d_u \log d_u)$.*

Theorem 8 (Certified robustness to injected profiles). *With frozen nuisances, one injected user v changes u 's 3-hop score of item k by $\Delta_k = x_k(D + a_k c_{vk}) - a_k x_k^2$, where $a = \tilde{w}_u$, $x = c_v \odot w_v$ and $D = \langle a, x \rangle$. If v logs at most L interactions with arbitrary items, labels and propensities $\geq \tau$, then D and x_k lie in closed-form intervals. Δ_k is linear in D and quadratic in x_k , so its extremes $[\underline{\Delta}_k, \overline{\Delta}_k]$ over this box are exact. Item t provably stays out of u 's top- K against any F such users if $s_{ut} + F\overline{\Delta}_t$ is below the $(K+1)$ -th largest value of $s_{uk} + F\underline{\Delta}_k$. With unclipped IPS, the influence of a single interaction is unbounded.*

Theorem 9 (Joint differential privacy). *Scale each user row so that $\|\tilde{w}_v\|_2 \leq R$ and $\|c_v^{\odot 2} \odot w_v\|_2 \leq R^2$. Releasing G plus symmetric Gaussian noise with $\sigma = 2R^2\sqrt{2\ln(1.25/\delta)}/\epsilon$ is (ϵ, δ) -DP under adding or removing one user. Scoring by (4) uses only the release and the user's own row, so the recommender is (ϵ, δ) -jointly DP (Hsu et al., 2014). Post-processing such as low-rank denoising is free.*

Theorem 10 (Exposure-capped allocation with a causal utility guarantee). *Let $\hat{x}$ maximise $\langle x, s \rangle$ over allocations with K items per user and at most cap_i users per item. This is a totally unimodular b -matching, solved exactly by min-cost flow up to a certified rounding gap Γ . Then every cap holds, at*

least $RK/\max_i \text{cap}_i$ items are recommended, and for the full-exposure utility $U(x) = \langle x, F^* \rangle$,

$$U(\hat{x}) \geq \max_x U(x) - \Gamma - 2 \max_x |\langle x, s - F^* \rangle|.$$

For DRUP, $\mathbb{E}s = F^*$, and Theorem 4(d) with a union bound makes the last term $O(RK t_\delta)$ with probability $1 - \delta$. For logged-graph propagation, $\mathbb{E}s$ is the exposure-weighted $F^*(P \odot Y)$, and the term does not vanish.

6. Experiments

6.1. Setup

Data. *Coat* (Schnabel et al., 2016) has 290 users and 300 items. Training consists of 6,960 self-selected (MNAR) ratings. Testing uses 16 ratings per user on random items (MAR), and we rank only items the user never rated. The dataset ships propensities; we also fit our own exposure model, with the same conclusions. *KuaiRec* (Gao et al., 2022) provides the big matrix (7,176 users, 10,728 items, 10.3M logged views) as the MNAR log and a fully observed small matrix (1,411 users, 3,327 items, 99.6% dense) as the test set. We rank all never-logged small-matrix items. *Yahoo!R3* (Marlin and Zemel, 2009) has 311,704 self-selected song ratings of 15,400 users on 1,000 songs (MNAR log, 2% dense) and 54,000 ratings of 10 uniformly random songs for 5,400 of the users (MAR test), in the version distributed with Chen et al. (2021). A random 5% of the MAR users (270) is held out entirely and used only to estimate $P(Y=1 \mid \text{MAR})$ for an outcome-dependent Naive-Bayes propensity (Schnabel et al., 2016). For every method the propensity model (logistic exposure model or Naive Bayes) is a hyper-parameter chosen

on validation data. Positives are ratings ≥ 4 (Coat, Yahoo!R3) and watch ratio ≥ 2 (KuaiRec).

Protocol. Hyper-parameters are chosen on a validation part of the unbiased data: 30% of each user’s test items for Coat, 30% of the test users for KuaiRec and Yahoo!R3. We report the held-out remainder, averaged over 10 (Coat) or 5 (KuaiRec, Yahoo!R3) random splits. Trained baselines are popularity, imputation only, and MF / IPS-MF / DR-MF / LightGCN / NAVIP-LightGCN / DR-LightGCN with 64-dimensional embeddings, point-wise losses and early stopping on the validation part. On Coat they use 10 splits and a grid over learning rate and weight decay. On KuaiRec the grid (learning rate $\{10^{-2}, 3 \cdot 10^{-3}\} \times$ weight decay $\{10^{-4}, 10^{-5}\}$, up to 30 epochs, patience 3) is searched on the validation part of the first split. The selected configuration is then trained on all 5 splits, the same splits DRUP uses, so every comparison is paired. The training-free propagation family shares one scoring form and differs only in its operator: the logged graph (linear LightGCN), IPS adjacency (NAVIP-style), IPS with the walk correction (WC), DR adjacency, and DRUP.

6.2. Monte-Carlo verification

Table 1 uses a synthetic MNAR model with known propensities: 20,000 replications for $K = 3$, 4,000 for $K = 5$, and 3,000 for the elasticity. At three hops the uncorrected IPS and DR operators have relative biases of 2.4 and 1.9. At five hops these grow to 78 and 70, because repeated walks multiply the inverse propensities. With the walk correction the bias vanishes. At $K = 3$ the fraction of entries with $|z| > 3$ matches the 0.27% expected by

Table 1: Monte-Carlo check of Theorems 1, 3 and 5.

Estimator	$K = 3$			$K = 5$	
	rel. —bias—	frac. $ z > 3$	rel. RMSE	rel. —bias—	elasticity η
Obs	0.992	1.0000	1.02	0.991	+1.777
IPS	2.403	0.4042	21.01	77.592	+0.007
IPS+WC	0.028	0.0025	4.93	0.106	+0.007
DR	1.903	0.6292	20.94	69.640	-0.187
DRUP	0.020	0.0008	3.97	0.140	+0.005

chance, and RMSE drops about fivefold. At $K = 5$ the residual 0.11–0.14 is within the Monte-Carlo standard error: the relative RMSE is about 12, so the standard error with 4,000 replications is about 0.19. The elasticity follows Theorem 5: +1.78 for the logged graph, -0.19 for DR adjacency, and ≈ 0 for DRUP. The K -hop engine reproduces the closed-form 3-hop operator to 7×10^{-15} and brute-force enumeration of all 5-hop walks to 10^{-13} .

6.3. Accuracy

Table 2 reports accuracy and Table 3 the paired t -tests over shared splits. On KuaiRec, DRUP with low-rank imputation needs no gradient training and reaches nDCG@20 of 0.6385. It is significantly better than every tuned trained baseline: MF 0.6320 ($p = 0.005$), NAVIP 0.6302 ($p = 0.005$), LightGCN 0.6300 ($p = 0.006$), IPS-MF 0.6287, DR-LightGCN 0.6247 and DR-MF 0.6191. It is also better than DRUP with the additive imputation (0.6326) and than the low-rank imputation used alone (0.5506). The gain therefore comes from propagating over the doubly robust graph, not from the imputation itself. Extending the propagation to five hops (Theorem 2) adds a

further +0.0018 (nDCG@20 0.6402; $p = 0.11$ against three hops), and the 5-hop variant beats every trained baseline with $p < 0.01$. As a sensitivity check, lowering the learning rate of MF, LightGCN and NAVIP to 10^{-3} raised the split-0 validation score only for MF (by 0.002), lowered it for LightGCN and NAVIP, and never raised the test score. On Coat, five hops neither help nor hurt. On Yahoo!R3, validation selects the IPS end of the control-variate family ($\lambda = 0$) for both DR and DRUP. The log is only 2% dense, so almost every edge of a DR graph would be imputed, and the imputation alone is below popularity (0.524 against 0.533). DRUP then coincides with IPS adjacency (nDCG@5 0.662) and is significantly better than logged-graph propagation (+0.0035, $p = 0.001$). It is far ahead of every tuned trained baseline: NAVIP 0.591, LightGCN 0.588, DR-LightGCN 0.561 and MF-type models about 0.51, all with $p < 10^{-3}$. The outcome-dependent Naive-Bayes propensity did not help any method on validation, except NAVIP by a small margin. On Coat, DRUP significantly outperforms MF, IPS-MF, DR-MF, LightGCN and NAVIP, and ties with DR-LightGCN. It is, however, significantly *below* the additive imputation model (-0.014) and the uncorrected DR adjacency (-0.006). With 290 users and 16 random ratings each, the additive imputation already captures most of the recoverable signal, and the low-rank imputation over-fits. Across both data sets the walk correction changes top- K accuracy only marginally relative to the same propagation without it (e.g., 0.6385 against 0.6384 on KuaiRec). What the correction buys is unbiasedness, exposure invariance and certifiability. As predicted by Theorem 3, IPS+WC coincides with IPS on unexposed candidates at three hops, because the factor $W_{ui} = 0$ hides the IPS bias.

6.4. Fairness

Table 4 tests Theorem 5 on real data. We thin the logged exposures of a random half of the items with probability $1/2$, which is an intervention with a known propensity change, and re-run each method. Logged-graph propagation demotes the thinned items strongly: their mean within-user rank shifts by -0.134 on KuaiRec and -0.058 on Coat. In other words, it punishes under-exposure. Clipped IPS demotes them less on KuaiRec (-0.029) and as much on Coat (-0.059), where clipping at $\tau = 0.2$ makes it almost identical to the logged graph. DRUP’s shift is $+0.0005 \pm 0.0008$ on KuaiRec and $+0.0006 \pm 0.011$ on Coat, i.e., zero within noise, as Theorem 5 predicts. The uncorrected DR adjacency is also close to zero on these data. Its over-correction ($\eta = -0.19$ in Table 1) scales with $(1 - p)/p \cdot \varepsilon^2$ (Theorem 3), and the validation-selected clip and an accurate imputation keep it small here. The exposure-conditional bias ECB (KuaiRec, where item quality is known from the fully observed matrix) drops from 0.830 (logged graph) and 0.571 (IPS) to 0.418 (DRUP), and to 0.285 with the low-rank imputation. Under the same intervention, DRUP with low-rank imputation shifts the thinned items by $+0.0077 \pm 0.0004$. This is 17 times smaller than the logged graph’s shift but not zero. The reason is that the low-rank model is re-fitted on the intervened log, so its nuisance depends on O , which violates Assumption 2. The additive imputation has far fewer parameters and stays within noise. The nDCG gap between men and women on Coat shrinks from 0.112 to 0.079 . On KuaiRec, the gap between active and inactive users is small for every method (≤ 0.013).

When does invariance hold? The role of τ . Theorem 5 is a statement about *expected scores* under Assumptions 1–2 and $\pi p \geq \tau$. Table 5 measures the score elasticity η directly, with nuisances frozen at their values on the original log. The logged graph has $\eta = +1.000 \pm 0.011$ (Yahoo! R3) and $+1.03$ (Coat), exactly as predicted. On Coat and KuaiRec, DRUP with its validation-selected clip is invariant ($\eta = -0.02$ on Coat). On Yahoo! R3 the log is 2% dense and the median logged propensity is only 0.05, so the validation-selected clip $\tau = 0.2$ binds for 81% of logged pairs. DRUP then behaves almost like the logged graph ($\eta = +0.95$). Lowering τ restores invariance: $\eta = +0.006 \pm 0.039$ at $\tau \leq 5 \cdot 10^{-4}$. It costs accuracy, though (nDCG@5 0.620 instead of 0.662), which is exactly the clipping bias–variance trade-off of Theorem 4. Two further caveats come from these runs. First, re-fitting the nuisances on the intervened log leaves a residual elasticity. Second, expected-score invariance does not imply rank invariance when the scores are sums of few, heavily skewed inverse-propensity terms: on Yahoo! R3 the median-sensitive rank shift stays at -0.04 even when $\eta \approx 0$. On the denser KuaiRec log, both vanish.

Limitation. Exposure invariance is a causal guarantee, not a distributional one. When true quality is concentrated, DRUP’s top- K lists are concentrated too, with higher Gini and PRU than the logged graph. The accuracy–concentration frontier (Table 6) makes this trade-off explicit. On Coat, DRUP reaches only 0.379 nDCG@5 under a Gini cap of 0.4, while the logged graph reaches 0.492. On KuaiRec, IPS keeps 0.575 at Gini ≤ 0.98 , while every DRUP configuration in our grid exceeds 0.99. The exposure-capped re-ranker below (Table 7) addresses this with a hard, certified guar-

antee.

6.5. Exposure-capped re-ranking

Table 7 applies the same exposure-capped allocation to every operator. The min-cost-flow solver returns the exact optimum: all caps hold, coverage is never below the guaranteed $RK/\text{max cap}$, and the certified rounding gap is negligible. A dual-subgradient solver we tried first left gaps of up to 35%, which the certificate exposed. On KuaiRec, where exposure is extremely concentrated, DRUP with low-rank imputation is the most accurate operator *at every cap*. For example, it reaches 0.260 against 0.228 (logged graph) and 0.219 (IPS) at $c = 10$, and 0.143 against 0.115 and 0.111 at $c = 2$. With the additive imputation, DRUP is only on par with the logged graph under caps, because its non-personalised one-hop term puts every user’s mass on the same items. Two caveats remain. First, at equal *Gini*, logged-graph and IPS propagation still keep slightly more accuracy: below the cap they spread exposure more evenly (e.g., IPS reaches 0.219 at Gini 0.80, DRUP-LR 0.204). Second, on Coat, whose random test items are already diverse, capping costs DRUP more accuracy than it costs the logged graph. On Yahoo! R3 exposure is not concentrated (Gini 0.27), and caps down to 1.5 times the fair share cost every operator at most 0.01. Under a hard exposure-cap policy, then, DRUP is the best choice on concentrated data, while Gini-targeted fairness still favours the correlation-based operators.

6.6. Transparency

Attributions recomputed against full re-propagation are exact up to floating-point precision: 1.3×10^{-15} on Coat (float64) and 3×10^{-6} on KuaiRec

(float32), for completeness as well as for additivity over random subsets (Table 8). Removing the interactions with the largest attributions moves the top item further down than random or similarity-based removal. This holds for every removal budget on KuaiRec and for up to five removals on Coat, and it shows that the attributions are faithful. The absolute rank changes are small, because top recommendations rest largely on population-level evidence. Theorem 7 also answers a question approximate explainers cannot: whether any counterfactual explanation exists at all. On Coat, 82% of the top recommendations cannot be overturned by removing any subset of the user’s own interactions. On KuaiRec the figure is 45%, and the minimal explanation has a median size of 15.5 interactions. These recommendations are driven by population-level evidence, and DRUP certifies that they are.

6.7. Accountability: certified robustness

Table 9 reports the attack. Empirically, the DR-based operators are the hardest to manipulate. On KuaiRec, 100 fake users (1.4% of the user base) never place the target in any real user’s top-20 under DR or DRUP, while they already do so for 2.7% (logged graph) and 8.9% (IPS) of users. At 200 fake users the rates are 13% for DR and DRUP, against 29% and 49%. On Coat, DRUP resists up to 10 fake users, while the logged graph and IPS are breached at 5–10.

The certificates of Theorem 8 were never violated, and brute-force enumeration confirms the bounds in float32 and float64. They are informative when catalogs are small. On Coat, DRUP is the only operator with non-trivial certificates: 100% of users are certified against one 21-interaction fake profile, 99.6% against two and 21% against five. The reason is that de-

degrees taken from the imputation make the fake user’s normalisation known, whereas degrees taken from W are attacker-controlled and void the certificate. On KuaiRec the worst-case certificate is vacuous even for $F = 1$. With 10,728 items, a profile of 51 interactions tailored to a single victim can in the worst case shift scores by more than the gap between a low-popularity target and the top-20 boundary. The realised attacks stay far below this worst case. Tighter certificates for large catalogs, for instance under bounded profile norms, are an open problem. At intermediate budgets the uncorrected DR adjacency with attacker-controlled degrees is sometimes harder to push than DRUP, because the fake user’s large DR weights inflate its own degree. That protection cannot be certified, because an attacker can design a profile that avoids it. On Yahoo! R3, where DRUP selects the IPS end, it inherits the larger $1/\tau$ weights of IPS: at $F = 100$ the target reaches 23% of users against 20% for the logged graph, and only the logged graph obtains non-trivial certificates (12% of users at $F = 1$). Robustness therefore favours the DR end of the family and the imputation-based degrees.

6.8. Privacy

Table 10 shows the privacy–utility curve. On KuaiRec (7,176 users), jointly DP DRUP reaches nDCG@20 of 0.613 at $\epsilon = 8$ and 0.629 at $\epsilon = 16$. The non-private value is 0.633, trained LightGCN reaches 0.629 and popularity 0.577. On Coat, with only 290 users, the same guarantee costs considerably more accuracy. Useful privacy needs a large user base, as expected for user-level DP. On Yahoo! R3 (15,400 users, 1,000 items) the jointly private recommender reaches 0.595 at $\epsilon = 16$ against 0.657 non-private.

7. Discussion and limitations

DRUP relies on unconfoundedness and on nuisances that are fixed or cross-fitted, the standard assumptions of IPS/DR estimation. When a flexible imputation is re-fitted on the data it is applied to, a small residual exposure sensitivity remains (Section 6.4). The exact correction holds for any number of hops, but the number of coincidence patterns grows with the Bell numbers (2,790 contractions at $K = 7$). The item-side fairness guarantee is causal (exposure invariance). Distributional targets need the exposure-capped re-ranker, and at equal Gini the correlation-based operators still keep slightly more accuracy. Robustness certificates are informative only for small catalogs. The privacy analysis treats the low-dimensional nuisances as public. Finally, the accuracy gain of the walk correction itself is small. Its value lies in the guarantees: beyond accuracy, DRUP turns a graph recommender into an object whose bias, fairness, explanations, robustness, exposure allocation and privacy can be stated and checked.

8. Implications

For research. Debiasing a graph recommender is not only a matter of re-weighting its loss or its edges. Once propagation spans several hops, the propagation operator itself must be made unbiased, and repeated walks are the precise place where standard estimators fail. Treating recommendation scores as walk polynomials of edge estimates also makes their fairness, explanations, robustness and privacy analysable in closed form. This offers a way to state and test trustworthiness claims rather than only measure them.

For practice. DRUP needs no gradient training and runs at the cost of one item Gram matrix, so it can serve as a strong, auditable baseline or as a production scorer. Its exact attributions show which of a user’s own interactions drive a recommendation, or certify that none can overturn it. The public-operator form lets a platform publish one privatised item operator while users score locally. The exposure-capped allocation enforces hard provider-side exposure limits with a certified optimality gap. The choice of the clip τ and of the control-variate weight should follow the density of the log. On very sparse logs, exact exposure invariance requires small clips and costs accuracy (Table 5).

9. Conclusion

We identified the repeated-walk bias of inverse-propensity and doubly robust graph propagation, gave its exact form and a lower bound, and removed it with an idempotent walk correction. The correction is exact for any number of hops and costs no more than the uncorrected operator. The resulting estimator is unbiased and edge-wise doubly robust for propagation over the full-exposure graph. Its structure yields bounds on variance and clipping bias, causal exposure invariance, exact and optimal explanations, certified robustness, joint differential privacy and certified exposure-capped allocation. On three benchmarks with unbiased test data, DRUP is at least as accurate as tuned trained baselines and significantly more accurate on two of them. Future work includes privatising the nuisance models, tighter certificates for large catalogs, and combining the walk correction with learned propagation weights.

References

- Chen, J., Dong, H., Qiu, Y., He, X., Xin, X., Chen, L., Lin, G., Yang, K., 2021. AutoDebias: Learning to debias for recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 21–30. doi:10.1145/3404835.3462919.
- Chen, J., Dong, H., Wang, X., Feng, F., Wang, M., He, X., 2023. Bias and debias in recommender system: A survey and future directions. *ACM Transactions on Information Systems* 41, 1–39. doi:10.1145/3564284.
- Dwork, C., Roth, A., 2014. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science* 9, 211–407.
- Efron, B., Stein, C., 1981. The jackknife estimate of variance. *The Annals of Statistics* 9, 586–596.
- Gao, C., Li, S., Lei, W., Chen, J., Li, B., Jiang, P., He, X., Mao, J., Chua, T.S., 2022. KuaiRec: A fully-observed dataset and insights for evaluating recommender systems, in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM), pp. 540–550.
- He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., Wang, M., 2020. LightGCN: Simplifying and powering graph convolution network for recommendation, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 639–648.

- Hsu, J., Huang, Z., Roth, A., Roughgarden, T., Wu, Z.S., 2014. Private matchings and allocations, in: Proceedings of the 46th Annual ACM Symposium on Theory of Computing (STOC), pp. 21–30.
- Islam, M.A., Faruk, A.S., Medya, S., Zheleva, E., 2026. Debiasing message passing to mitigate popularity bias in GNN-based collaborative filtering. [arXiv:2605.11145](#).
- Jia, J., Liu, Y., Hu, Y., Gong, N.Z., 2023. PORE: Provably robust recommender systems against data poisoning attacks, in: Proceedings of the 32nd USENIX Security Symposium, pp. 1703–1720.
- Kim, M., Oh, J., Do, J., Lee, S., 2022. Debiasing neighbor aggregation for graph neural network in recommender systems, in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM). doi:10.1145/3511808.3557576.
- Lucic, A., ter Hoeve, M.A., Tolomei, G., de Rijke, M., Silvestri, F., 2022. CF-GNNExplainer: Counterfactual explanations for graph neural networks, in: Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS), pp. 4499–4511.
- Marlin, B.M., Zemel, R.S., 2009. Collaborative prediction and ranking with non-random missing data, in: Proceedings of the Third ACM Conference on Recommender Systems (RecSys), pp. 5–12.
- McDiarmid, C., 1989. On the method of bounded differences, in: Surveys in Combinatorics. Cambridge University Press, pp. 148–188.

- McSherry, F., Mironov, I., 2009. Differentially private recommender systems: Building privacy into the Netflix prize contenders, in: Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 627–636.
- Saito, Y., Yaginuma, S., Nishino, Y., Sakata, H., Nakata, K., 2020. Unbiased recommender learning from missing-not-at-random implicit feedback, in: Proceedings of the 13th International Conference on Web Search and Data Mining (WSDM), pp. 501–509.
- Schnabel, T., Swaminathan, A., Singh, A., Chandak, N., Joachims, T., 2016. Recommendations as treatments: Debiasing learning and evaluation, in: Proceedings of the 33rd International Conference on Machine Learning (ICML), pp. 1670–1679.
- Shen, Y., Wu, Y., Zhang, Y., Shan, C., Zhang, J., Letaief, K.B., Li, D., 2021. How powerful is graph convolution for recommendation?, in: Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM), pp. 1619–1629.
- Wang, X., Zhang, R., Sun, Y., Qi, J., 2019. Doubly robust joint learning for recommendation on data missing not at random, in: Proceedings of the 36th International Conference on Machine Learning (ICML), pp. 6638–6647.
- Wang, Y., Ma, W., Zhang, M., Liu, Y., Ma, S., 2023. A survey on the fairness of recommender systems. *ACM Transactions on Information Systems* 41, 52:1–52:43. doi:10.1145/3547333.

- Ying, R., Bourgeois, D., You, J., Zitnik, M., Leskovec, J., 2019. GNNExplainer: Generating explanations for graph neural networks, in: Advances in Neural Information Processing Systems (NeurIPS).
- Yu, D., Li, Q., Soundararaj, B., Pettit, C., Xu, G., 2026. A survey of causal learning in graph-based recommender systems. ACM Computing Surveys 58, 404:1–404:32. doi:10.1145/3840748.
- Zhang, Y., Chen, X., 2020. Explainable recommendation: A survey and new perspectives. Foundations and Trends in Information Retrieval 14, 1–101.
- Zheng, Y., Gao, C., Li, X., He, X., Li, Y., Jin, D., 2021. Disentangling user interest and conformity for recommendation with causal embedding, in: Proceedings of the Web Conference (WWW), pp. 2980–2991.

Appendix A. Proofs

Theorem 1.. After the correction every walk contributes $\prod_{e \in \text{distinct}} C_e^{k_e} W_e$. The W_e are independent (Assumption 1, fixed nuisances), so $\mathbb{E} \prod W_e = \prod \mathbb{E} W_e$ with $\mathbb{E} W_e = \hat{Y}_e + p_e(Y_e - \hat{Y}_e)/\bar{p}_e = Y_e + \delta_e$. The target monomial is $\prod C_e^{k_e} Y_e^{k_e} = \prod C_e^{k_e} Y_e$ because $Y_e \in \{0, 1\}$. $\square$

Theorem 3.. The naive operator contains, for $v = u, j \neq i$, the terms $\tilde{W}_{uj}^2 \tilde{W}_{ui}$; for $j = i, v \neq u$, the terms $\tilde{W}_{ui} \tilde{W}_{vi}^2$; and $\tilde{W}_{ui}^3$. Conditionally on $O_{ui} = 0$ we have $W_{ui} = \hat{Y}_{ui}$. By independence, $\mathbb{E}[W_e^2] = Y_e + \text{Var}(W_e)$ with $\text{Var}(W_e) = \frac{1-p_e}{p_e}(Y_e - \hat{Y}_e)^2$, while the corrected replacement has mean Y_e . Summing the excesses gives the formula. The lower bound uses $(1-p)/p \geq 1/\max p - 1$. $\square$

Theorem 4.. (a) $\text{Var}(W_e) = p_e(1 - p_e)\varepsilon_e^2/\bar{p}_e^2$. If $p_e \geq \tau$ this is at most $(1/\tau - 1)\varepsilon_e^2$; otherwise it is at most $p_e\varepsilon_e^2/\tau^2 < \varepsilon_e^2/\tau$. (b) $\hat{T}_{ui}$ is multilinear in independent variables and therefore affine in each W_e . The Efron–Stein inequality (Efron and Stein, 1981) gives $\text{Var}(\hat{T}) \leq \sum_e \mathbb{E}[(\partial_e \hat{T})^2] \text{Var}(W_e)$. The derivative $\partial_e \hat{T}$ is a sum, over the walks through e , of C -weights times at most two other independent W_f with $\|W_f\|_2^2 \leq 1 + 1/\tau$. Minkowski’s inequality then bounds $\|\partial_e \hat{T}\|_2 \leq (1 + 1/\tau)\Lambda_e$. (c) By Theorem 1, the bias is at most $\sum |\kappa_{\text{walk}}| |\prod(Y_f + |\delta_f|) - \prod Y_f| \leq ((1 + \bar{\delta})^3 - 1)M_{ui}$, and $|\delta_e| = (1 - p_e/\tau)_+ \varepsilon_e$. (d) Changing O_e moves W_e by at most ε_e/τ , and $|\partial_e \hat{T}| \leq B^2 \Lambda_e$. Apply McDiarmid’s inequality (McDiarmid, 1989). $\square$

Theorem 5.. DRUP: by Theorem 1, $\mathbb{E}[s] = F^*$, which does not depend on π . Logged graph: $\mathbb{E}[O_e Y_e] = p_e Y_e$, so each 3-hop monomial containing $k \geq 1$ edges incident to i scales as π^k . Uncorrected DR: by Theorem 3, $\mathbb{E}[s] = F^* + C_{ui} \hat{Y}_{ui} K_i(\pi) + \dots$, and $K_i(\pi)$ is strictly decreasing in π . $\square$

Proposition 6 and Theorem 7.. Substitute $G = G^{(-u)} + \tilde{w}_u^\top \tilde{w}_u - \text{diag}(\tilde{w}_u^2) + \text{diag}(c_u^2 w_u)$ into (4). The terms $\tilde{w}_{ui} \sum_j \tilde{w}_{uj}^2$ cancel against the local correction, leaving $\tilde{w}_{ui} \sum_j c_{uj}^2 w_{uj}$, and every remaining term involving w_{uj} ($j \neq i$) is linear. Additive games have Shapley values equal to their individual effects. For Theorem 7, the margin after removing S is $s_{ui} - s_{uk} - \sum_{j \in S} g_j$, so minimising $|S|$ under $\sum_S g_j > \text{margin}$ selects the largest g_j . $\square$

Theorem 8.. Adding row v adds contrib_v to G and leaves u ’s local terms unchanged. Then $\Delta = \tilde{w}_u \text{contrib}_v = (a \cdot x)x - a \odot x^2 + a \odot c_v \odot x$. Logged DR entries satisfy $w \in [\hat{Y} - \hat{Y}/\tau, \hat{Y} + (1 - \hat{Y})/\tau]$ for any label and any $p \geq \tau$, and unlogged ones equal $\hat{Y}$. D is maximised (minimised) by assigning the L

slots to the largest positive (negative) deviations. For fixed x_k the function is linear in D , so it attains its extremes at the ends of the D interval. For fixed D it is a quadratic in x_k , whose extremes over an interval lie at the ends or at the clamped vertex. Effects add over fake users. For unclipped IPS, $|W_e| = 1/p_e$ is unbounded. $\square$

Theorem 9. $\|\text{contrib}_v\|_F \leq \|\tilde{w}_v \tilde{w}_v^\top - \text{diag}(\tilde{w}_v^2)\|_F + \|c_v^2 w_v\|_2 \leq R^2 + R^2$, so the ℓ_2 -sensitivity of the upper triangle is at most $2R^2$, and the Gaussian mechanism (Dwork and Roth, 2014) applies. Each user's output is a function of the DP release and their own data, which gives joint DP by the billboard lemma (Hsu et al., 2014). $\square$

Table 2: Accuracy on unbiased test data (mean $\pm$ std over splits). Best in bold, second underlined. The KuaiRec 5-hop rows use the low-rank imputation.

Method	metrics (Coat, Yahoo! R3: nDCG@5, Recall@5, nDCG@10; KuaiRec: nDCG@20, Recall@20, nDCG@50)		
<i>Coat</i>			
Popularity	0.5367±0.0095	0.6441±0.0093	0.6599±0.0086
Imputation only	0.5662±0.0095	0.6736±0.0120	0.6787±0.0071
MF	0.4619±0.0224	0.5666±0.0213	0.6056±0.0178
IPS-MF	0.4569±0.0234	0.5579±0.0241	0.6049±0.0178
DR-MF	0.5357±0.0141	0.6473±0.0162	0.6575±0.0113
LightGCN	0.5077±0.0173	0.6036±0.0216	0.6403±0.0109
NAVIP-LightGCN	0.4994±0.0199	0.5979±0.0246	0.6329±0.0129
DR-LightGCN	0.5560±0.0241	0.6565±0.0254	0.6725±0.0165
Linear LightGCN (logged graph)	0.5571±0.0100	0.6461±0.0094	<u>0.6745±0.0076</u>
IPS adjacency (NAVIP-style)	0.5438±0.0186	0.6403±0.0152	0.6641±0.0138
IPS adjacency + WC	0.5438±0.0186	0.6403±0.0152	0.6641±0.0138
DR adjacency	<u>0.5586±0.0124</u>	<u>0.6728±0.0143</u>	0.6737±0.0088
DRUP (ours)	0.5526±0.0134	0.6637±0.0157	0.6720±0.0108
DR adjacency, 5 hops	0.5551±0.0131	0.6683±0.0132	0.6720±0.0105
DRUP, 5 hops (ours)	0.5545±0.0082	0.6681±0.0130	0.6725±0.0095
<i>KuaiRec</i>			
Popularity	0.5766±0.0023	0.1656±0.0016	0.4610±0.0019
Imputation only	0.6215±0.0014	0.1867±0.0020	0.5112±0.0013
MF	0.6320±0.0041	0.1925±0.0022	0.5101±0.0050
IPS-MF	0.6287±0.0028	0.1913±0.0018	0.5077±0.0024
DR-MF	0.6191±0.0020	0.1878±0.0016	0.5044±0.0026
LightGCN	0.6300±0.0028	0.1927±0.0025	0.5114±0.0016
NAVIP-LightGCN	0.6302±0.0025	0.1921±0.0024	0.5135±0.0028
DR-LightGCN	0.6247±0.0027	0.1917±0.0014	0.5070±0.0018
Linear LightGCN (logged graph)	0.6036±0.0028	0.1815±0.0016	0.4985±0.0024
IPS adjacency (NAVIP-style)	0.5954±0.0022	0.1829±0.0015	0.4892±0.0017
IPS adjacency + WC	0.5954±0.0022	0.1829±0.0015	0.4892±0.0017
DR adjacency	0.6318±0.0017	0.1927±0.0023	0.5144±0.0019
DRUP (ours)	0.6326±0.0018	0.1929±0.0025	0.5147±0.0015
DR adjacency, 5 hops	0.6404±0.0015	0.1941±0.0021	<u>0.5201±0.0017</u>
DRUP, 5 hops (ours)	<u>0.6402±0.0012</u>	<u>0.1941±0.0021</u>	0.5201±0.0016
DR adjacency, low-rank $\hat{Y}$	0.6384±0.0028	0.1941±0.0024	0.5194±0.0027
DRUP, low-rank $\hat{Y}$ (ours)	0.6385±0.0028	0.1941±0.0024	0.5194±0.0027
<i>Yahoo! R3</i>			
Popularity	0.5331±0.0059	0.6794±0.0044	0.6556±0.0042
Imputation only	0.5241±0.0047	0.6875±0.0046	0.6447±0.0036
MF	0.5070±0.0056	0.6630±0.0063	0.6347±0.0043
IPS-MF	0.5115±0.0049	0.6674±0.0073	0.6381±0.0029
DR-MF	0.5086±0.0033	0.6744±0.0061	0.6341±0.0034
LightGCN	0.5876±0.0049	0.7210±0.0071	0.6929±0.0030
NAVIP-LightGCN	0.5913±0.0070	0.7246±0.0090	0.6957±0.0042
DR-LightGCN	0.5608±0.0040	0.7077±0.0037	0.6733±0.0042
Linear LightGCN (logged graph)	<u>0.6585±0.0058</u>	<u>0.7809±0.0067</u>	<u>0.7434±0.0035</u>
IPS adjacency (NAVIP-style)	0.6620±0.0064	0.7849±0.0066	0.7453±0.0040
IPS adjacency + WC	0.6620±0.0064	0.7849±0.0066	0.7453±0.0040
DR adjacency	0.6620±0.0064	0.7849±0.0066	0.7453±0.0040
DRUP (ours)	0.6620±0.0064	0.7849±0.0066	0.7453±0.0040

Table 3: Paired t -tests of DRUP against each baseline (nDCG@5 on Coat, nDCG@20 on KuaiRec).

Data	comparison	splits	mean diff.	p (paired t)
Coat	DRUP vs Pop	10	+0.0159	0.0021
Coat	DRUP vs Impute	10	-0.0136	0.000517
Coat	DRUP vs Obs	10	-0.0044	0.356
Coat	DRUP vs IPS	10	+0.0088	0.249
Coat	DRUP vs IPS+WC	10	+0.0088	0.249
Coat	DRUP vs DR	10	-0.0060	0.00272
Coat	DRUP vs MF	10	+0.0908	8.78e-08
Coat	DRUP vs IPS-MF	10	+0.0957	5.91e-07
Coat	DRUP vs DR-MF	10	+0.0170	0.000639
Coat	DRUP vs LightGCN	10	+0.0449	2.08e-06
Coat	DRUP vs NAVIP	10	+0.0533	3.06e-05
Coat	DRUP vs DR-LightGCN	10	-0.0033	0.635
KuaiRec	DRUP-LR vs Pop	5	+0.0619	2.23e-07
KuaiRec	DRUP-LR vs Impute	5	+0.0170	7.68e-05
KuaiRec	DRUP-LR vs Obs	5	+0.0348	9.72e-07
KuaiRec	DRUP-LR vs IPS	5	+0.0431	1.72e-07
KuaiRec	DRUP-LR vs IPS+WC	5	+0.0431	1.72e-07
KuaiRec	DRUP-LR vs DR	5	+0.0067	0.00233
KuaiRec	DRUP-LR vs DRUP (additive $\hat{Y}$)	5	+0.0059	0.00664
KuaiRec	DRUP-LR vs Impute-LR	5	+0.0879	6.25e-08
KuaiRec	DRUP-LR vs MF	5	+0.0065	0.00464
KuaiRec	DRUP-LR vs IPS-MF	5	+0.0098	0.000462
KuaiRec	DRUP-LR vs DR-MF	5	+0.0194	8.98e-05
KuaiRec	DRUP-LR vs LightGCN	5	+0.0085	0.00569
KuaiRec	DRUP-LR vs NAVIP	5	+0.0082	0.00549
KuaiRec	DRUP-LR vs DR-LightGCN	5	+0.0137	0.000886
KuaiRec	DRUP-5hop-LR vs MF	5	+0.0083	0.00518
KuaiRec	DRUP-5hop-LR vs IPS-MF	5	+0.0116	0.000172
KuaiRec	DRUP-5hop-LR vs DR-MF	5	+0.0212	6.07e-06

Table 4: Fairness. ECB is the partial correlation between an item’s mean rank and its log exposure, controlling for its true like rate. The user gap is the nDCG gap between user groups (gender on Coat, activity on KuaiRec). The last column is the rank shift of items whose exposures were thinned by a real $do(p \leftarrow p/2)$; 0 means exposure invariance.

Data	Method	nDCG	ECB↓	PRU↓	Gini↓	user gap↓	rank shift under $do(p/2)$
Coat	Obs	0.4917	0.769	0.394	0.340	0.1123	-0.0585±0.0073
Coat	IPS	0.4864	0.765	0.392	0.337	0.1189	-0.0594±0.0066
Coat	DR	0.4770	0.744	0.763	0.655	0.0795	-0.0027±0.0122
Coat	DRUP	0.4803	0.739	0.758	0.655	0.0793	+0.0006±0.0110
KuaiRec	Obs	0.6026	0.830	0.845	0.991	0.0080	-0.1342±0.0009
KuaiRec	IPS	0.5950	0.571	0.756	0.983	0.0023	-0.0290±0.0011
KuaiRec	DR	0.6324	0.418	0.831	0.994	0.0116	+0.0006±0.0007
KuaiRec	DRUP	0.6334	0.418	0.831	0.994	0.0129	+0.0005±0.0008
KuaiRec	DRUP-LR	0.6395	0.285	0.793	0.994	0.0136	+0.0077±0.0004
Yahoo! R3	Obs	0.6590	0.856	0.534	0.281	0.0457	-0.0943±0.0018
Yahoo! R3	IPS	0.6616	0.851	0.520	0.274	0.0470	-0.0890±0.0017
Yahoo! R3	DR	0.6616	0.851	0.520	0.274	0.0470	-0.0890±0.0017
Yahoo! R3	DRUP	0.6616	0.851	0.520	0.274	0.0470	-0.0890±0.0017

Table 5: Clip τ versus exposure invariance under a real $do(p \leftarrow p/2)$ on a random half of the items, with nuisances frozen (Assumption 2). η is the elasticity of the mean score of treated relative to control items (1: exposure-proportional, 0: invariant).

Data	operator	τ	nDCG	score elasticity η	rank shift
Coat	Obs	–	0.4917	+1.034 $\pm$ 0.043	-0.0646
Coat	DRUP	0.2	0.4803	-0.020 $\pm$ 0.005	+0.0061
Coat	DRUP	0.05	0.4839	-0.069 $\pm$ 0.018	+0.0188
Coat	DRUP	0.01	0.4188	-0.094 $\pm$ 0.060	+0.0212
Coat	DRUP	0.001	0.4107	-0.044 $\pm$ 0.057	+0.0121
KuaiRec	Obs	–	0.6026	+0.997 $\pm$ 0.003	-0.2412
KuaiRec	DRUP	0.1	0.6305	-0.013 $\pm$ 0.000	+0.0035
KuaiRec	DRUP	0.02	0.3393	+0.001 $\pm$ 0.004	+0.0027
KuaiRec	DRUP	0.001	0.1942	-0.045 $\pm$ 0.022	+0.0030
Yahoo! R3	Obs	–	0.6590	+1.000 $\pm$ 0.011	-0.1604
Yahoo! R3	DRUP	0.2	0.6616	+0.954 $\pm$ 0.012	-0.1507
Yahoo! R3	DRUP	0.05	0.6551	+0.814 $\pm$ 0.025	-0.1282
Yahoo! R3	DRUP	0.02	0.6385	+0.652 $\pm$ 0.022	-0.1070
Yahoo! R3	DRUP	0.005	0.6248	+0.254 $\pm$ 0.006	-0.0630
Yahoo! R3	DRUP	0.002	0.6195	+0.060 $\pm$ 0.011	-0.0468
Yahoo! R3	DRUP	0.001	0.6205	+0.012 $\pm$ 0.035	-0.0440
Yahoo! R3	DRUP	0.0005	0.6204	+0.006 $\pm$ 0.039	-0.0435
Yahoo! R3	DRUP	0.0002	0.6204	+0.006 $\pm$ 0.039	-0.0435

Table 6: Best nDCG reachable by each operator under a cap on the Gini coefficient of top- K exposure (over the whole hyper-parameter grid).

Data	Method	Gini cap			
		≤ 0.4	≤ 0.5	≤ 0.6	none
Coat	Obs	0.4917	0.4917	0.4917	0.4917
Coat	IPS	0.4864	0.4864	0.4864	0.4864
Coat	DR	0.4090	0.4487	0.4733	0.4853
Coat	DRUP	0.3786	0.4416	0.4848	0.4902
		≤ 0.97	≤ 0.98	≤ 0.99	none
KuaiRec	Obs	0.3543	0.3543	0.3543	0.6026
KuaiRec	IPS	0.5398	0.5745	0.5950	0.5950
KuaiRec	DR	–	0.3394	0.4633	0.6324
KuaiRec	DRUP	–	–	–	0.6328

Table 7: Exposure-capped re-ranking (Theorem 10, exact min-cost flow). Each item may be recommended to at most c times its fair share RK/N of users. Cells: nDCG@ K (Gini@ K). Every cap was satisfied and every certified rounding gap was below 10^{-4} .

<i>Coat</i> , cap factor c	none	20	10	5	3	2	1.5
Obs	0.502 (0.34)	0.494 (0.34)	0.501 (0.34)	0.501 (0.34)	0.493 (0.33)	0.494 (0.32)	0.487 (0.28)
IPS	0.499 (0.33)	0.486 (0.34)	0.491 (0.33)	0.496 (0.35)	0.492 (0.34)	0.484 (0.32)	0.477 (0.28)
DR	0.477 (0.65)	0.477 (0.65)	0.477 (0.65)	0.477 (0.65)	0.470 (0.64)	0.453 (0.51)	0.441 (0.39)
DRUP	0.480 (0.66)	0.480 (0.66)	0.480 (0.66)	0.480 (0.66)	0.473 (0.64)	0.445 (0.51)	0.441 (0.39)

<i>KuaiRec</i> , cap factor c	none	50	20	10	5	3	2
Obs	0.603 (0.99)	0.425 (0.96)	0.300 (0.91)	0.228 (0.84)	0.173 (0.72)	0.139 (0.60)	0.115 (0.44)
IPS	0.595 (0.98)	0.411 (0.94)	0.292 (0.87)	0.219 (0.80)	0.166 (0.70)	0.134 (0.58)	0.111 (0.44)
DR	0.632 (0.99)	0.386 (0.98)	0.285 (0.95)	0.228 (0.90)	0.179 (0.80)	0.147 (0.67)	0.121 (0.50)
DRUP	0.633 (0.99)	0.386 (0.98)	0.285 (0.95)	0.230 (0.90)	0.179 (0.80)	0.147 (0.67)	0.120 (0.50)
DRUP-LR	0.639 (0.99)	0.457 (0.98)	0.332 (0.95)	0.260 (0.90)	0.204 (0.80)	0.169 (0.67)	0.143 (0.50)

<i>Yahoo! R3</i> , cap factor c	none	20	10	5	3	2	1.5
Obs	0.658 (0.28)	0.659 (0.28)	0.660 (0.28)	0.658 (0.28)	0.656 (0.28)	0.658 (0.27)	0.652 (0.23)
IPS	0.660 (0.27)	0.663 (0.27)	0.661 (0.27)	0.662 (0.27)	0.661 (0.27)	0.659 (0.27)	0.654 (0.22)
DR	0.660 (0.27)	0.663 (0.27)	0.661 (0.27)	0.662 (0.27)	0.661 (0.27)	0.659 (0.27)	0.654 (0.22)
DRUP	0.660 (0.27)	0.663 (0.27)	0.661 (0.27)	0.662 (0.27)	0.661 (0.27)	0.659 (0.27)	0.654 (0.22)

Table 8: Exact explanations (Prop. 6, Thm. 7) on 200 random test users.

Data	max rel. error	mean min. CF size	median	users without CF
Coat	1.3e-15	2.36	2	0.82
KuaiRec	3.3e-06	24.40	16	0.45
Yahoo! R3	6.7e-16	3.11	1	0.18

Table 9: Injection attack: F fake users log a low-popularity target item and the most popular items. Left: realised hit rate of the target in real users’ top- K . Right: fraction of real users for whom Theorem 8 certifies that the target cannot enter the top- K against *any* F profiles of that size. No certificate was violated.

Coat: $K=5$, fake profile size 21														
	hit rate of target in top- K ↓							certified fraction ↑						
F	0	1	2	5	10	20	50	0	1	2	5	10	20	50
Obs	0.004	0.004	0.004	0.009	0.108	0.444	0.661	0.996	0.000	0.000	0.000	0.000	0.000	0.000
IPS	0.004	0.004	0.004	0.020	0.201	0.502	0.651	0.995	0.000	0.000	0.000	0.000	0.000	0.000
DR	0.000	0.000	0.000	0.000	0.000	0.000	0.655	1.000	0.000	0.000	0.000	0.000	0.000	0.000
DRUP	0.000	0.000	0.000	0.000	0.000	0.382	0.973	1.000	1.000	0.996	0.211	0.000	0.000	0.000

KuaiRec: $K=20$, fake profile size 51																
	hit rate of target in top- K ↓								certified fraction ↑							
F	0	1	2	5	20	50	100	200	0	1	2	5	20	50	100	200
Obs	0.002	0.002	0.002	0.002	0.002	0.002	0.027	0.288	0.998	0.000	0.000	0.000	0.000	0.000	0.000	0.000
IPS	0.002	0.002	0.002	0.002	0.002	0.005	0.089	0.485	0.998	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DR	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.135	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DRUP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.132	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Yahoo! R3: $K=5$, fake profile size 21																
	hit rate of target in top- K ↓								certified fraction ↑							
F	0	1	2	5	10	20	50	100	0	1	2	5	10	20	50	100
Obs	0.065	0.065	0.065	0.065	0.065	0.065	0.071	0.199	0.935	0.117	0.047	0.000	0.000	0.000	0.000	0.000
IPS	0.065	0.065	0.065	0.065	0.065	0.065	0.080	0.234	0.935	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DR	0.065	0.065	0.065	0.065	0.065	0.065	0.080	0.234	0.935	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DRUP	0.065	0.065	0.065	0.065	0.065	0.065	0.080	0.234	0.935	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 10: nDCG of jointly DP DRUP ($\delta = 10^{-5}$). The denoising rank is chosen on validation data.

Data	$\epsilon=0.5$	1	2	4	8	16	∞ (clipped)	non-private / Pop
Coat	0.4094	0.4118	0.4182	0.4651	0.5068	0.5301	0.5398	0.5326 / 0.5226
KuaiRec	0.0536	0.0826	0.1473	0.5569	0.6127	0.6290	0.6346	0.6325 / 0.5766
Yahoo! R3	0.3662	0.3630	0.3858	0.4576	0.5360	0.5950	0.6451	0.6566 / 0.5253